

# FUNDX

## Systematic Multi-Asset Funds with News-Sentiment Analytics

### FINS3645 FinTech Project 2026 - Part B

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Live app <https://z5544937-fundx.streamlit.app>

Public repository [https://github.com/1Tehan/z5544937\\_projectB](https://github.com/1Tehan/z5544937_projectB)

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#### PART B IN ONE SENTENCE

FundX is now a tested product rather than a concept: 17 walk-forward funds, a sector news-sentiment index, a finance-specific VADER extension, a look-ahead-safe fusion test, explicit trading costs, a block-bootstrap robustness check, and a deployed seven-page investor app.

Out-of-sample fund design | sentiment measurement | investor-facing implementation

# Executive Result

## PRODUCT DECISION

Lead with Combined Risk Parity for a balanced investor. It does not earn the highest raw return, but it has the strongest combined-fund Sharpe (0.862), a materially smaller drawdown than Combined Maximum Sharpe (-19.5% versus -22.7%), and only 3.7% average monthly turnover. Keep Minimum Variance as the defensive option, Equal Weight as the transparent benchmark, and treat standalone crypto as specialist/high-risk exposure.

- The simple benchmark remains hard to beat: Equity Equal Weight records a 0.784 Sharpe and outperforms every optimised equity-only fund on this OOS sample.
- Crypto helps only when sizing controls its risk. Combined Equal Weight is worse than Equity Equal Weight on Sharpe (0.701 versus 0.784), while Combined Risk Parity is better than Equity Risk Parity (0.862 versus 0.682).
- Fin-VADER improves sentiment measurement but not investment performance: neutral classification falls from 49.6% to 45.2%, yet the Fin-VADER tilted fund has a 0.342 Sharpe versus 0.380 for the untilted base.
- The 21-day block bootstrap prevents overclaiming. Every selected fund's 90% Sharpe interval spans zero, so the final product choice uses drawdown and turnover as well as the point-estimate Sharpe.

## 1. The Funds and the Backtest Design

Part A defined FundX for an investor who wants diversification and a clear rule but does not want to select individual securities or accept an opaque allocation. Part B turns that proposition into an investable shelf. Each FundX portfolio has a named construction rule, a fact sheet with out-of-sample performance and current holdings, and a place in the app where the user can compare funds and combine them into an allocation. The original fee-on-assets idea is also implemented in the allocation builder as an illustrative 0.75% annual fee.

The core shelf contains fifteen funds: equity-only, crypto-only and combined equity-plus-crypto, each built with five allocation methods. Two additional equity funds apply the sentiment tilt, taking the total to seventeen. The methods are intentionally different in what they estimate and what role they play in the product:

| Method                                          | Role in FundX                       | Main estimation exposure                  |
|-------------------------------------------------|-------------------------------------|-------------------------------------------|
| Equal Weight                                    | Transparent 1/N benchmark           | None beyond the investable universe       |
| Minimum Variance                                | Defensive / lowest-risk option      | Covariance estimates                      |
| Maximum Sharpe                                  | Growth-oriented optimiser           | Expected returns + covariance             |
| Risk Parity (ERC; Spinu, 2013)                  | Balanced risk-budgeting option      | Risk/covariance, no return forecast       |
| Hierarchical Risk Parity (López de Prado, 2016) | Robust clustering-based alternative | Dependence structure, no matrix inversion |

All funds use a genuine walk-forward backtest. At each monthly rebalance, on the first trading day of the month, the target weights are estimated only from observations strictly before that date. Equity and combined funds use a 252-trading-day window and become live on 4 January 2021; crypto-only funds use 365 calendar days and become live on 1 January 2021. Equity-calendar results are annualised by 252 and crypto-only results by 365, with a zero risk-free rate for Sharpe. The OOS sample ends on 29 December 2023 for equity/combined funds and 31 December 2023 for crypto-only funds.

The fund engine is long-only and fully invested. Minimum Variance and Maximum Sharpe explicitly enforce a 20% per-asset cap for equity and combined funds and a 40% cap for crypto-only funds. Equal Weight is below those thresholds by construction; I leave ERC and HRP unclipped because post-hoc clipping would change their defining allocation rules, and their realised OOS weights remain below the same thresholds. For numerical stability, Minimum Variance, Maximum Sharpe and ERC shrink the covariance matrix 10% toward its diagonal. The sentiment tilt instead clips its sector multiplier to 0.6-1.4 and does not re-impose a final stock-level cap. Target weights drift with realised returns between rebalances, so turnover is measured against the portfolio that actually arrives at the next rebalance. I charge 10 basis points multiplied by turnover on rebalance days and report net-of-cost Sharpe alongside gross Sharpe.

The three fund families are reported separately because they answer different investor questions and use different calendars. Equity-only funds test whether an allocation rule adds value inside a conventional large-cap universe. Crypto-only funds isolate a much higher-volatility asset class on its seven-day calendar. Combined funds are the closest match to FundX's diversification proposition because they ask whether crypto can add something useful once it competes with equities for portfolio risk. I therefore compare methods within each family first and then compare like-for-like methods

across families, rather than letting the 2021 crypto rally dominate the conclusion. The explicit caps on Minimum Variance and Maximum Sharpe are product guardrails against concentration from noisy estimates; ERC and HRP are evaluated in their native form. This keeps the backtest focused on rules a retail product can disclose and repeat while being precise about which constraints each method actually uses.

#### BACKTEST GUARDRAIL

I also ran a falsification check: planting a +60% return on a rebalance date did not change that date's target weights. Separately, I recomputed Combined Risk Parity from fund\_returns.csv and matched the saved annualised return, volatility, Sharpe, maximum drawdown and growth-of-\$1 metrics.

## 2. Out-of-Sample Results and Fund Fact Sheets

The most useful way to read the leaderboard is as a product decision rather than a race for the highest return. Combined Maximum Sharpe finishes with the highest combined-family wealth (\$1.61 from \$1), but Combined Risk Parity is the stronger balanced fund because it produces the best risk-adjusted return with much lower trading intensity. Table 1 reports all 17 funds; Figures 1-5 show the growth, family comparison, drawdown, changing weights and cross-method Sharpe ranking.

| Selected fund         | Ann. ret. | Ann. vol. | Sharpe | Max DD | Growth \$1 | Turnover |
|-----------------------|-----------|-----------|--------|--------|------------|----------|
| Combined Risk Parity  | 14.0%     | 16.2%     | 0.862  | -19.5% | \$1.48     | 3.7%     |
| Combined Max Sharpe   | 17.3%     | 23.2%     | 0.744  | -22.7% | \$1.61     | 28.5%    |
| Combined Min Variance | 5.5%      | 12.6%     | 0.434  | -15.5% | \$1.17     | 13.0%    |
| Equity Equal Weight   | 12.6%     | 16.1%     | 0.784  | -20.2% | \$1.43     | 2.6%     |
| Crypto Min Variance   | 59.8%     | 71.6%     | 0.836  | -72.0% | \$4.08     | 10.2%    |
| Crypto Max Sharpe     | -3.0%     | 76.0%     | -0.040 | -85.6% | \$0.91     | 21.8%    |

Combined Risk Parity earns 14.0% annualised with 16.2% volatility and a 0.862 Sharpe. Its maximum drawdown is -19.5%, \$1 grows to \$1.48, and average monthly turnover is only 3.7%, so net Sharpe barely moves to 0.859. Figure 5 shows it as the strongest point-estimate Sharpe across the broad FundX set. I implement equal-risk-contribution weights using the Spinu (2013) log-barrier formulation. The important mechanism is sizing: the combined fund can use crypto as a separate source of returns, but ERC prevents the much more volatile crypto assets from receiving the same capital weight as an equity merely because both are in the universe.

That mechanism is clearest in the equal-weight comparison. Equity Equal Weight has a 0.784 Sharpe; Combined Equal Weight falls to 0.701 even though annualised return rises from 12.6% to 15.1%. Volatility rises more, from 16.1% to 21.6%, because ten crypto assets receive the same nominal weight as each of the fifty equities. In contrast, Combined Risk Parity improves sharply on Equity Risk Parity (0.862 versus 0.682). In this sample, the question is not simply whether to add crypto; it is whether the sizing rule prevents crypto volatility from dominating the portfolio.

The strongest benchmark result sits inside the equity family. Equity Equal Weight beats every optimised equity-only fund on Sharpe: 0.784 versus 0.682 for risk parity, 0.611 for HRP, 0.436 for minimum variance and 0.380 for maximum Sharpe. This is consistent with the 1/N result in DeMiguel, Garlappi and Uppal (2009): out-of-sample estimation error can erase the theoretical gain from optimisation. Maximum Sharpe is the clearest example because it depends on expected returns as well as covariance estimates, and expected returns are especially unstable over a 252-day window. Equal Weight's edge is also partly structural: it estimates nothing and trades least (2.6%), avoiding both estimation error and turnover drag. A stricter comparison would hold turnover fixed across methods. For FundX, complexity is therefore not a product benefit unless it survives the benchmark.

Minimum Variance does what its label promises. Both the combined and equity versions have about 12.6% annualised volatility and the smallest drawdowns in their families (-15.5% and -15.3%), but only 5.5% annualised return. This gives FundX a defensible low-risk option even though it is not the growth leader. HRP follows López de Prado (2016): it clusters the dependence structure and allocates recursively rather than inverting the covariance matrix. Combined HRP has 14.0% volatility, a -17.9% drawdown and a 0.701 Sharpe. Its contribution is a different, less inversion-dependent construction method rather than spectacular outperformance.

The crypto-only family is the clearest warning against ranking a fund by return in isolation. Crypto Minimum Variance earns 59.8% annualised and grows \$1 to \$4.08, yet still experiences a -72.0% maximum drawdown. Crypto Equal Weight draws down -81.6% and Crypto HRP -78.1%. Crypto Maximum Sharpe is worse again: -3.0% annualised return, a gross Sharpe of -0.040 (net -0.043 after the 10 bps turnover cost), \$0.91 ending wealth and an -85.6% drawdown. Figure 2 shows the 2021 boom and the 2022 collapse. These funds can be shown in FundX, but a standalone crypto fund is a specialist exposure, not the default expression of a diversified-retail proposition.

Combined Maximum Sharpe is still important because it exposes the trade-off between return, concentration and implementation. It earns 17.3% annualised and ends at \$1.61, but volatility is 23.2%, Sharpe is 0.744 and maximum drawdown reaches -22.7%. Figure 3 places the deepest trough on 27 September 2022. Figure 4 shows the source of concentration: the 20% cap binds repeatedly, and at the final 1 December 2023 rebalance the five largest target weights in fund\_weights.csv are GE 20.0%, NVDA 19.4%, Southern Company (SO) 16.3%, Adobe (ADBE) 12.4% and BTC-USD 10.3%. Together they are 78.4% of the target portfolio. The cap prevents even more extreme bets, but it does not make the optimiser diversified in the everyday sense of the word.

The 2021-2023 window also contains a useful stress test rather than only favourable market conditions. Combined Maximum Sharpe reaches a wealth peak of roughly \$1.73 in April 2022 and then experiences its deepest drawdown later that year; by the end of the sample it has not fully recovered that prior peak. Risk Parity gives up some terminal wealth but avoids part of that path dependence. This matters for a retail product because two funds with similar long-run narratives can create very different experiences for a user who invests near a local peak. The results therefore support a shelf rather than a single universal winner: Minimum Variance is defensible when loss control is the priority, Equal Weight is valuable because it is transparent and unexpectedly competitive, Maximum Sharpe can be shown as a higher-concentration growth option, and Risk Parity is the strongest balanced default on this sample. I would not use the standalone crypto results as headline marketing despite their spectacular annualised returns. A -72% to -86% drawdown range means the user could lose most of the displayed account value before any long-run average return becomes relevant. The combined-family evidence is more consistent with FundX's original promise of intelligent diversification.

#### INTERPRETATION RULE

Figure 1 and Figure 5 deliberately tell different stories: Maximum Sharpe finishes with the most money in the combined family, while Risk Parity has the better Sharpe. FundX should not collapse those two questions into a single "best fund" label.

### 3. The Standalone Sentiment Index

The sentiment pipeline continues the Part A data rules rather than rebuilding them. The same src/etl.py is carried into Part B: headlines are deduplicated on ticker, date and title, removing 2,847 exact duplicates and leaving 146,836 headlines. UTC-aware dates are normalised and news is mapped to the same or next equity trading day. I keep the raw headline text because VADER uses punctuation, emphasis and local word context as part of its rules (Hutto and Gilbert, 2014).

Each headline receives a compound score. I first average headlines to ticker-day sentiment, then equal-weight the tickers with news into a sector-day index. Ticker-days with no headlines are dropped rather than filled with zero: zero is a genuine VADER output meaning approximately balanced tone, while no headline is missing data. Zero-filling would mechanically pull thin-coverage sectors toward zero and create false stability. The standalone index is descriptive; when sentiment enters a trading decision in src/fusion.py, it is lagged by at least one trading day, so information aligned to day t is not usable until day t+1.

Figure 6 shows that the 21-day sector indices are mostly above zero. Plain VADER's mean headline score is +0.106; 37.4% of headlines are positive, 13.0% negative and 49.6% neutral. The broad index weakens around the March 2020 COVID shock - the same period in which Part A verified OXY falling 52% on 9 March - but it does not become deeply negative. That is economically informative: the index contains publisher and corporate-news selection effects as well as market stress. Utilities also sit persistently above many other sectors, but I treat that as a descriptive pattern rather than evidence that Utilities genuinely had the strongest underlying economic news because coverage and headline mix differ by sector.

The index is therefore most credible as a relative, time-varying news measure rather than a literal "investor mood" gauge. It uses headlines rather than article bodies, and general-language VADER leaves a large zero spike. Those two limitations motivate the Fin-VADER extension and the later fusion test, but neither limitation is assumed away.

A second validation point is the consistency with Part A. Part A predicted, before the Part B scoring was run, that roughly half of the headlines would be neutral under general-purpose VADER. The measured 49.6% neutral share is therefore not a post-hoc story invented after seeing the result; it confirms a limitation that had already been identified from the vocabulary analysis. At the same time, the positive mean should not be read as proof that the market was continuously optimistic. Corporate and financial-news feeds disproportionately publish announcements, upgrades, launches and other event language, and sectors with fewer headlines can be influenced by a smaller set of stories. For that reason, I use the index primarily to compare changes through time and relative sector tone. The fusion experiment then provides the stricter test: if the descriptive signal contains investable information, a lagged implementation should improve the portfolio on at least one economically relevant dimension. It does not, which is why the report separates 'better sentiment measurement' from 'better investment performance.'

## 4. Extensions, Innovation and What the Evidence Changed

Innovation is not concentrated in one feature. The project adds a newer portfolio method (HRP), a finance-specific sentiment lexicon, a look-ahead-safe fusion experiment, explicit turnover costs, a block-bootstrap robustness test, and a consistent FundX design system. The important part is not the number of additions; it is that each one produces an observable result that changes how I would build or describe the product.

| Extension        | What was implemented                                            | Evidence / decision                                                                   |
|------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------|
| HRP              | Clustering-based allocation alongside four conventional methods | Adds a distinct, lower-volatility construction rule; does not beat the best benchmark |
| Fin-VADER        | ~90 curated finance terms/phrases added to VADER                | Neutral share 49.6% -> 45.2%; measurement improves, performance does not              |
| Sentiment fusion | Lagged 21-day sector z-score tilt with bounded multipliers      | Sharpe 0.380 -> 0.342; do not market it as alpha                                      |
| Turnover costs   | 10 bps x drift-adjusted turnover at rebalances                  | Penalises Max Sharpe more than Risk Parity; rankings mostly unchanged                 |
| Block bootstrap  | 21-day blocks, 2,000 draws, fixed seed                          | Sharpe intervals are wide; product decision should not rely on rank alone             |
| Design system    | Shared navy/teal/gold visual language across plots and app      | Makes evidence and product read as one FundX system                                   |

### 4.1 Fin-VADER: improving the measurement instrument

Part A predicted that plain VADER would leave roughly half of finance headlines neutral. The real Part B result is 49.6%, so the problem is visible before the extension is evaluated. Fin-VADER adds roughly ninety finance-specific words and phrase mappings on VADER's -4 to +4 valence scale. Examples include upgrade, downgrade, profit warning, all-time high, bankruptcy and weak demand. AI proposed candidate terms, but the shipping list and valences were reviewed and curated rather than treated as automatic ground truth; the scores remain hand-set rather than estimated from returns.

| Model                        | Neutral | Mean   | Std. dev. | Positive | Negative |
|------------------------------|---------|--------|-----------|----------|----------|
| Plain VADER                  | 49.6%   | +0.106 | 0.288     | 37.4%    | 13.0%    |
| Fin-VADER (extended lexicon) | 45.2%   | +0.117 | 0.312     | 40.6%    | 14.2%    |

Figure 7 shows the measurement effect: neutral share falls from 49.6% to 45.2%, about 6,500 headlines move out of the neutral bucket, mean sentiment rises from 0.106 to 0.117 and standard deviation rises from 0.288 to 0.312. Both the positive and negative shares increase. The extension therefore does exactly what it was designed to do - it recovers direction from finance language that plain VADER treats as neutral - without implying that the new direction is automatically tradable.

### 4.2 Fusion: a negative result that changes the product claim

The fusion rule is intentionally transparent. I lag the sector index one trading day, take a 21-day trailing mean, calculate a cross-sectional z-score across the ten sectors, multiply each base stock weight by  $\text{clip}(1 + 0.25 z_{\text{sector}}, 0.6, 1.4)$ , and renormalise. The lag removes look-ahead, the z-score measures relative rather than absolute tone, and the clip prevents an unbounded sector bet. All variants use the same Equity Maximum-Sharpe engine and OOS dates, so Table 2 is a like-for-like test.

| Table 2. Fusion comparison<br>OOS 2021-01-04 to 2023-12-29 | Ann. ret. | Ann. vol. | Sharpe | Max DD | Turnover | Net Sharpe |
|------------------------------------------------------------|-----------|-----------|--------|--------|----------|------------|
| Base Equity Max Sharpe                                     | 6.6%      | 17.3%     | 0.380  | -22.6% | 28.1%    | 0.359      |
| + VADER tilt                                               | 6.2%      | 17.9%     | 0.344  | -22.0% | 29.9%    | 0.323      |
| + Fin-VADER tilt                                           | 6.1%      | 17.8%     | 0.342  | -21.4% | 30.0%    | 0.320      |

The tilt does not improve the base. Sharpe falls from 0.380 to 0.344 under plain VADER and to 0.342 under Fin-VADER; turnover rises from 28.1% to roughly 30% per month. Maximum drawdown improves modestly from -22.6% to -21.4% under Fin-VADER, but Figure 8 shows that the three wealth paths are almost indistinguishable and the tilted funds finish slightly lower. A direct predictive check is similarly weak: across 9,800 sector-day observations, the Pearson information coefficient is -0.043 and the Spearman coefficient is -0.023 - both near zero and economically negligible. The most plausible explanation is not one single failure. Headline tone is noisy at a monthly horizon, the base Maximum-Sharpe fund is already weak out of sample, the signal is sector-level while weights are stock-level, and the 21-day aggregation smooths much of the headline-level difference that Fin-VADER created. On this evidence, FundX should expose sentiment as analytics and research, not advertise the tilt as an alpha feature.



### 4.3 Turnover, costs and implementation realism

Turnover separates the methods almost as clearly as Sharpe. Equal Weight trades only 2.6%-5.8% of portfolio value per month on average across families and Risk Parity 3.1%-6.3%, while Maximum Sharpe reaches 28.5% in the combined family and 28.1% in equity. At 10 bps the cost model does not overturn the ranking, but it lands exactly where expected: Combined Maximum Sharpe falls from 0.744 gross Sharpe to 0.727 net, whereas Combined Risk Parity barely changes from 0.862 to 0.859. Figure 9 shows the wealth lines staying close over three years, which is itself useful - the assumption is not large enough to manufacture a result - but the drag would scale with trading costs and is structurally highest in the most aggressive optimiser.

### 4.4 Robustness: the leaderboard is less certain than the point estimates

A 21-day moving-block bootstrap with 2,000 draws and seed 42 changes how I interpret the fund ranking. The same resampled date blocks are applied to every selected fund within each draw, preserving cross-fund dependence for paired comparisons. Figure 10 reports 90% intervals for selected Sharpe ratios. Every interval spans zero: Combined Risk Parity is 0.862 with approximately [-0.25, 2.02], while Combined Maximum Sharpe is 0.744 with approximately [-0.37, 1.61]. The paired Risk-Parity-minus-Max-Sharpe difference is +0.118 with a 90% interval of about [-0.41, 0.95], and Risk Parity is better in 70.9% of draws. The Fin-VADER tilt is worse than its base in 70.6% of paired draws, with a paired interval that also crosses zero. Three OOS years are therefore enough to reveal drawdowns, turnover and a failed fusion effect, but not enough to claim that small Sharpe differences are persistent. This is why the product recommendation uses several dimensions rather than point-estimate rank alone.

Taken together, the extensions form an evidence chain. HRP broadens the portfolio baseline; costs test implementation drag; Fin-VADER asks whether the text instrument can be improved; fusion tests whether that improvement reaches portfolio outcomes, and the near-zero information coefficient shows that the signal itself - not only the tilt mechanics - lacks predictive content at this horizon; the bootstrap then tests whether the apparent fund ranking is robust. The sequence matters because an extension can succeed at one layer and fail at the next. Fin-VADER improves measurement but not Sharpe, and the bootstrap prevents small point-estimate differences becoming false certainty. I therefore treat the negative fusion result as an innovation outcome rather than hiding it: the contribution is the disciplined evidence showing what changed, what did not, and how that should alter the FundX product claim.

## 5. The App and the Investor Journey

The deployed Streamlit app is where the analysis becomes an investor journey. It has seven pages: Overview, Compare funds, Fund fact sheet, Find my mix, Build your allocation, Sentiment analytics, and Methodology & data. The sequence is deliberate: a user can understand the shelf before choosing a fund, inspect one fund before allocating money, and still reach the underlying methodology instead of treating the model as a black box.

| Journey step   | What the user can do                                                             | Why it matters                                                   |
|----------------|----------------------------------------------------------------------------------|------------------------------------------------------------------|
| 1. Discover    | Overview + Find my mix introduce the shelf and risk preferences                  | Reduces a 17-fund universe to a manageable starting point        |
| 2. Compare     | Filter funds, sort metrics and overlay growth of \$1                             | Makes return, risk and drawdown trade-offs visible side by side  |
| 3. Inspect     | Open a fact sheet with KPIs, drawdown, holdings and weight history               | Turns each rule into an explainable investable product           |
| 4. Allocate    | Combine funds, normalise to 100%, enter amount/horizon and view fee illustration | Connects the backtest to an actual portfolio decision            |
| 5. Investigate | Switch VADER/Fin-VADER, change smoothing and inspect methodology/coverage        | Keeps sentiment and model limitations visible rather than hidden |

The allocation builder also makes the Part A business model concrete. An equal one-third allocation to Combined Maximum Sharpe, Combined Risk Parity and Equity Minimum Variance produces about 12.6% annualised return, 15.6% volatility, a 0.81 Sharpe and -18.9% maximum drawdown over the common OOS history. On a \$10,000 illustrative investment, the 0.75% annual fee displays as \$75 for the first year. These are historical, not promised, outcomes; the point of the feature is to make the consequence of a user's allocation visible.

The deployment architecture is intentionally light. Streamlit reads precomputed files under results/ and does not run VADER, NLTK or an optimiser at runtime. Heavy computation is performed by scripts/run\_part\_b.py, while source data is accessed through src/data\_access.py and is not committed to the public repository. This keeps the free-tier deployment responsive and makes the app reproducible on a basic machine. The report and app use the same FundX navy/teal/gold design language so the analytical evidence and the product interface feel like the same system.

The app also acts as a communication control on the analysis. A user does not see only a Sharpe leaderboard: the comparison page exposes return, volatility, drawdown and growth together, while the fact sheet reveals holdings and the history of target weights. That is particularly important for Maximum Sharpe, where the final five positions account for 78.4% of the target portfolio despite the 20% cap. The 'Find my mix' and allocation pages then shift the decision from 'which fund won?' to 'what combination matches the user's risk preference and amount invested?' Sentiment analytics

are deliberately placed alongside coverage and methodology rather than presented as a buy/sell signal. This mirrors the empirical conclusion that sentiment is informative as an analytic but not yet validated as alpha. The product design therefore reflects the evidence hierarchy in the report: historical performance is visible, assumptions are inspectable, and experimental features are not disguised as promises. That transparency is central to the target user, who wants help with portfolio construction without handing the entire decision to an unexplained black box.

#### LIVE PRODUCT

App: <https://z5544937-fundx.streamlit.app> | Public repo: [https://github.com/1Tehan/z5544937\\_projectB](https://github.com/1Tehan/z5544937_projectB)

## 6. Critical Reflection and Three Recommendations

The main lesson from Part B is that the most sophisticated-looking result is not necessarily the most useful product result. I expected optimisation to dominate the simple benchmark once the walk-forward backtest was implemented correctly. It did not. Equity Equal Weight beats every optimised equity fund on Sharpe, and Maximum Sharpe combines unstable expected-return estimates with the highest turnover and visible concentration. Risk Parity emerged as the better FundX lead because it solves a simpler problem - equalising risk contribution - and in this sample that simplicity survives out of sample better.

The second lesson is that measurement quality and investment alpha are different questions. Fin-VADER clearly improves the instrument: the neutral bucket shrinks by 4.4 percentage points and the score distribution carries more direction. Yet the portfolio does not improve, and the near-zero information coefficient confirms that the signal itself - not just the fusion mechanics - lacks predictive content at this horizon. That negative result is useful because it stops FundX from turning a descriptive sentiment feature into a performance claim that the evidence does not support.

The third lesson is that a retail product has to be judged by the loss experience, not only by average return. A 59.8% annualised return sounds extraordinary until it is placed next to a -72.0% drawdown. The crypto-only results therefore change the product shelf: crypto can remain available, but the evidence supports risk-budgeted combined exposure far more strongly than a standalone default.

There are also three material limitations. First, the OOS sample is only three years, and the bootstrap confirms that Sharpe uncertainty is wide. Second, the cost model is a simple linear 10 bps assumption; it does not model spreads, market impact or costs changing through stressed markets. Third, the sentiment system sees headlines rather than full articles and Fin-VADER valences are curated rather than statistically estimated. These limitations do not invalidate the results, but they define the next tests that matter.

The recommendations are ordered by that same principle. Data quality comes first because a single bad return can contaminate every covariance estimate and every fund built on top of it. Sentiment calibration comes second because the current experiment shows a clear measurement change but no portfolio benefit, so the next useful question is whether the hand-set valences can be replaced by evidence from returns. Turnover control comes third because it directly targets the largest implementation weakness of Maximum Sharpe and can be evaluated against the existing gross-versus-net framework. I would not prioritise adding more optimisation methods before these three steps. The project already shows that a larger method menu does not guarantee a better OOS product, while the simple benchmark and Risk Parity remain competitive. The more defensible development path is therefore to strengthen the data, estimate experimental parameters more rigorously, and make the optimisation objective closer to the return the customer actually receives. That keeps future FundX development tied to failures and frictions observed in this backtest rather than to novelty for its own sake.

### Recommendation 1 - Productionise the Part A extreme-move screen

Part A retained 29 observations with  $|z| > 8$  after verification. The next version should turn that screen into a live data-quality monitor that flags a new extreme price move before it can silently affect returns, covariance estimates and fund weights. This is more valuable than adding another optimiser because it protects every downstream model and closes the loop on an extension already identified in Part A.

### Recommendation 2 - Estimate, rather than hand-set, finance valences

Fin-VADER proves that finance vocabulary changes classification, but the current valences remain curated judgements and the near-zero information coefficient shows that the resulting signal has little predictive content at this horizon. The next version should regress next-day abnormal returns on term-presence dummies for the roughly 90 lexicon words, estimate the model on 2020-2022, and evaluate it on a held-out 2023 test sample. I would retain only terms whose direction and statistical evidence survive out of sample, then rerun the same fusion rule without retuning it. This directly tests whether the vocabulary is useful but the hand-set score magnitudes are not.

### Recommendation 3 - Put turnover inside the optimiser

Combined Maximum Sharpe turns over 28.5% of the portfolio per month versus 3.7% for Combined Risk Parity. A production version should add either an explicit turnover penalty or no-trade bands around existing positions and optimise net-of-cost performance directly. This aligns the objective with what a customer actually receives instead of optimising a gross statistic and subtracting implementation costs later.

#### FINAL FUNDX POSITION

FundX should lead with Combined Risk Parity for balanced growth, retain Minimum Variance for defensive users and Equal Weight as the benchmark, keep crypto-only funds clearly high-risk, and present sentiment as an analytic until a stronger fusion test demonstrates incremental value.

## References

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## Appendix A - Required Exhibits and Supporting Evidence

The appendix contains the required Part B exhibits plus the block-bootstrap robustness figure. Every exhibit is generated from the committed real-data results and is referenced and interpreted in the written narrative. The full performance table includes net Sharpe as an additional implementation metric.

| Fund                                           | Ann. ret. | Ann. vol. | Sharpe | Max DD | Growth \$1 | Turnover | Net Sharpe |
|------------------------------------------------|-----------|-----------|--------|--------|------------|----------|------------|
| Combined Equal Weight                          | 15.1%     | 21.6%     | 0.701  | -27.9% | \$1.52     | 3.8%     | 0.698      |
| Combined Minimum Variance                      | 5.5%      | 12.6%     | 0.434  | -15.5% | \$1.17     | 13.0%    | 0.421      |
| Combined Maximum Sharpe                        | 17.3%     | 23.2%     | 0.744  | -22.7% | \$1.61     | 28.5%    | 0.727      |
| Combined Risk Parity (ERC)                     | 14.0%     | 16.2%     | 0.862  | -19.5% | \$1.48     | 3.7%     | 0.859      |
| Combined Hierarchical Risk Parity              | 9.8%      | 14.0%     | 0.701  | -17.9% | \$1.32     | 10.3%    | 0.691      |
| Equity-only Equal Weight                       | 12.6%     | 16.1%     | 0.784  | -20.2% | \$1.43     | 2.6%     | 0.782      |
| Equity-only Minimum Variance                   | 5.5%      | 12.6%     | 0.436  | -15.3% | \$1.17     | 12.8%    | 0.423      |
| Equity-only Maximum Sharpe                     | 6.6%      | 17.3%     | 0.380  | -22.6% | \$1.21     | 28.1%    | 0.359      |
| Equity-only Risk Parity (ERC)                  | 9.9%      | 14.5%     | 0.682  | -18.5% | \$1.33     | 3.1%     | 0.679      |
| Equity-only Hierarchical Risk Parity           | 8.4%      | 13.7%     | 0.611  | -17.4% | \$1.27     | 11.9%    | 0.600      |
| Crypto-only Equal Weight                       | 40.5%     | 81.9%     | 0.495  | -81.6% | \$2.77     | 5.8%     | 0.493      |
| Crypto-only Minimum Variance                   | 59.8%     | 71.6%     | 0.836  | -72.0% | \$4.08     | 10.2%    | 0.833      |
| Crypto-only Maximum Sharpe                     | -3.0%     | 76.0%     | -0.040 | -85.6% | \$0.91     | 21.8%    | -0.043     |
| Crypto-only Risk Parity (ERC)                  | 44.3%     | 79.9%     | 0.555  | -79.9% | \$3.01     | 6.3%     | 0.553      |
| Crypto-only Hierarchical Risk Parity           | 43.2%     | 76.9%     | 0.562  | -78.1% | \$2.94     | 10.4%    | 0.559      |
| Equity Max Sharpe + Sentiment Tilt (VADER)     | 6.2%      | 17.9%     | 0.344  | -22.0% | \$1.20     | 29.9%    | 0.323      |
| Equity Max Sharpe + Sentiment Tilt (Fin-VADER) | 6.1%      | 17.8%     | 0.342  | -21.4% | \$1.19     | 30.0%    | 0.320      |

Table 1. Fact-sheet performance metrics for all 17 funds. Equity and combined funds: OOS 2021-01-04 to 2023-12-29, annualisation x252. Crypto-only funds: OOS 2021-01-01 to 2023-12-31, annualisation x365.  $r_f = 0$ . Net Sharpe uses 10 bps x turnover.

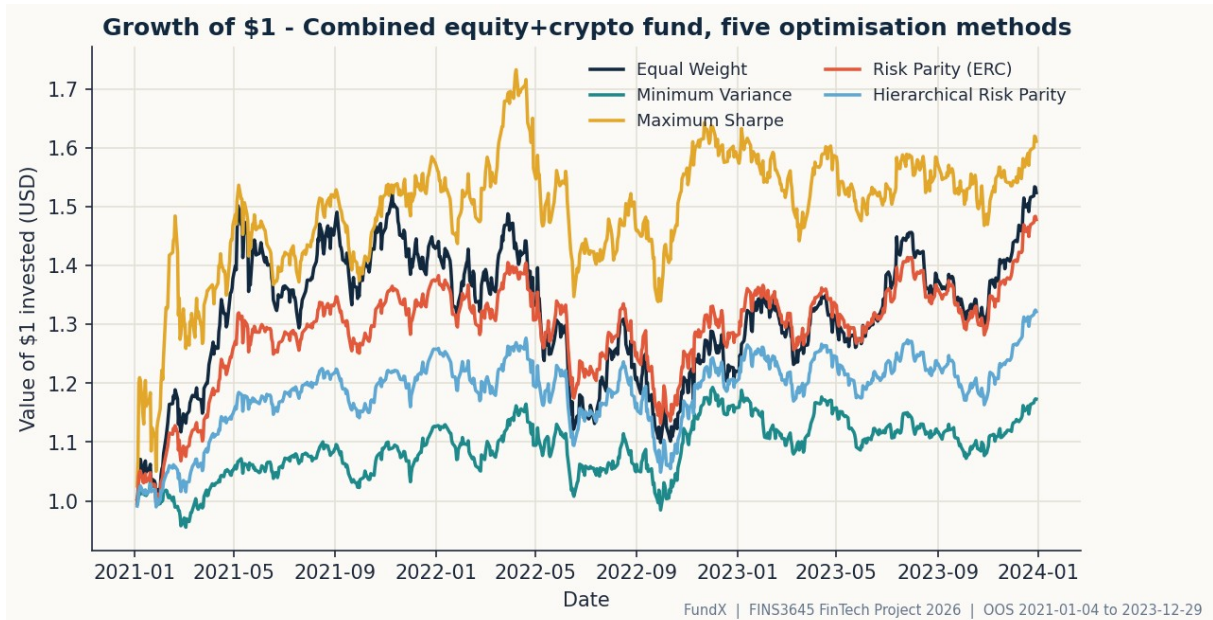


Figure 1. Growth of \$1 for the combined equity+crypto family under five allocation methods. Units: USD value of \$1 invested; OOS 2021-01-04 to 2023-12-29.

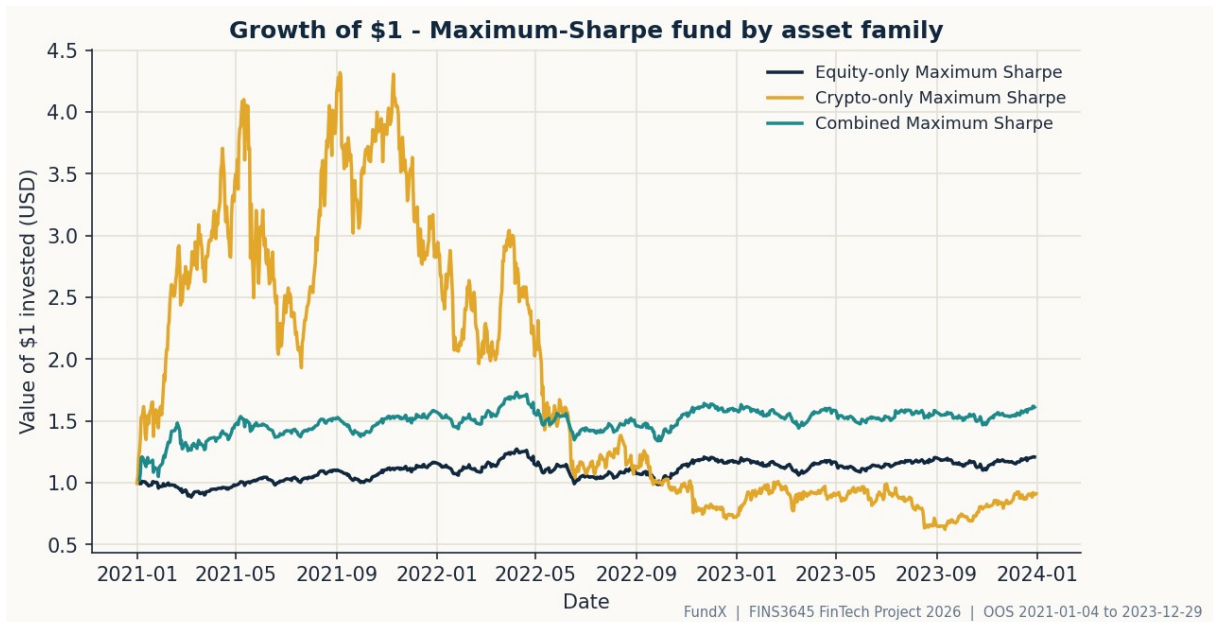


Figure 2. Growth of \$1 for Maximum-Sharpe funds by asset family. Units: USD value of \$1 invested; equity/combined OOS 2021-01-04 to 2023-12-29, crypto-only OOS 2021-01-01 to 2023-12-31.



Figure 3. Drawdown from the prior wealth peak for Combined Maximum Sharpe. Units: percentage below prior peak; OOS 2021-01-04 to 2023-12-29.

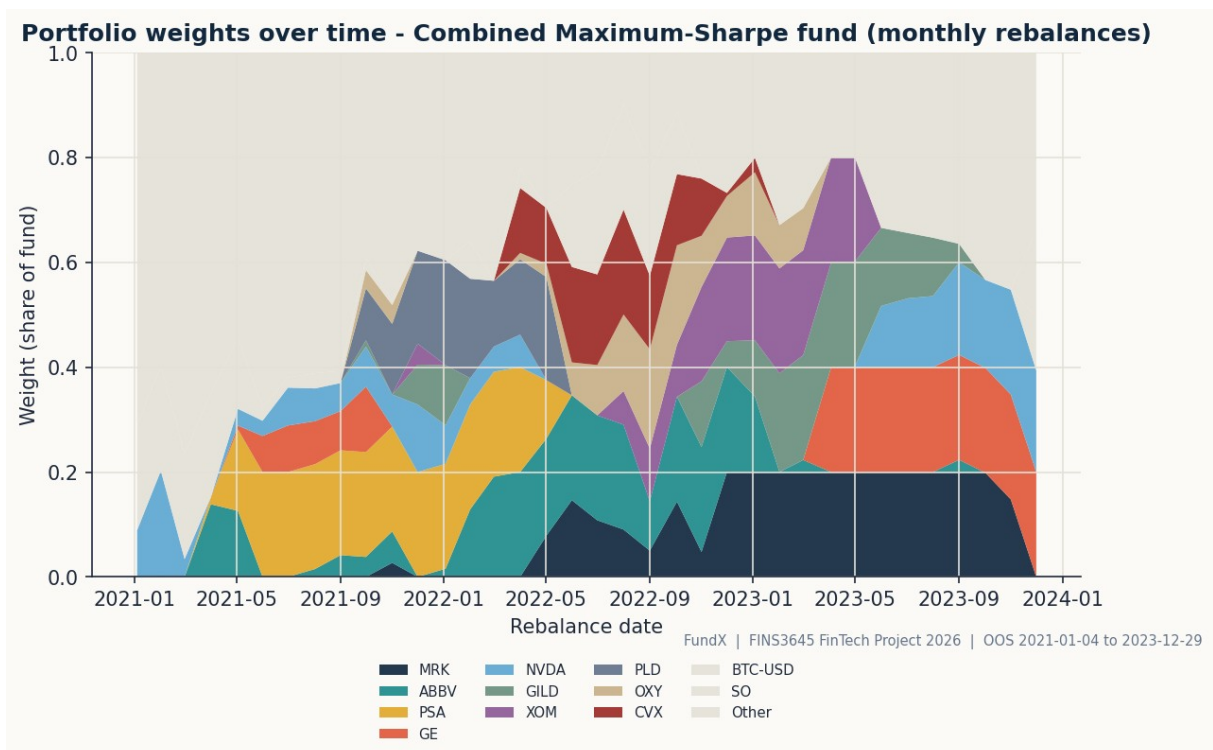


Figure 4. Target weights at monthly rebalances for Combined Maximum Sharpe. Units: share of fund; explicit 20% per-asset cap; OOS rebalances 2021-01-04 to 2023-12-01.

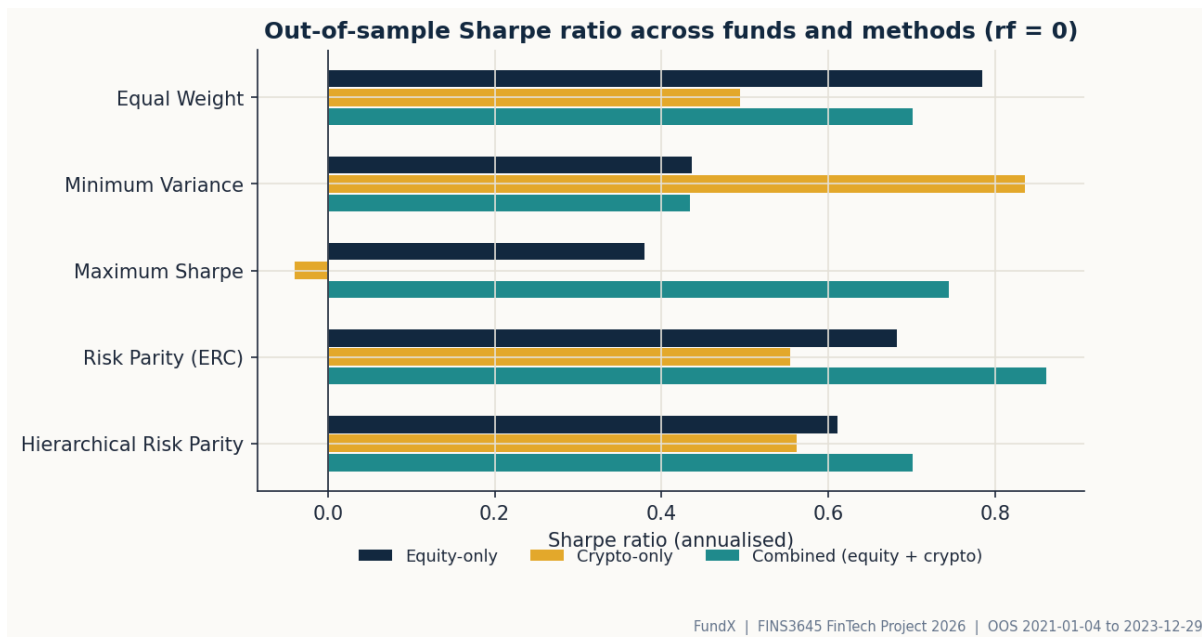


Figure 5. Annualised out-of-sample Sharpe ratios across fund families and allocation methods,  $rf = 0$ . Equity/combined OOS 2021-01-04 to 2023-12-29; crypto-only OOS 2021-01-01 to 2023-12-31.

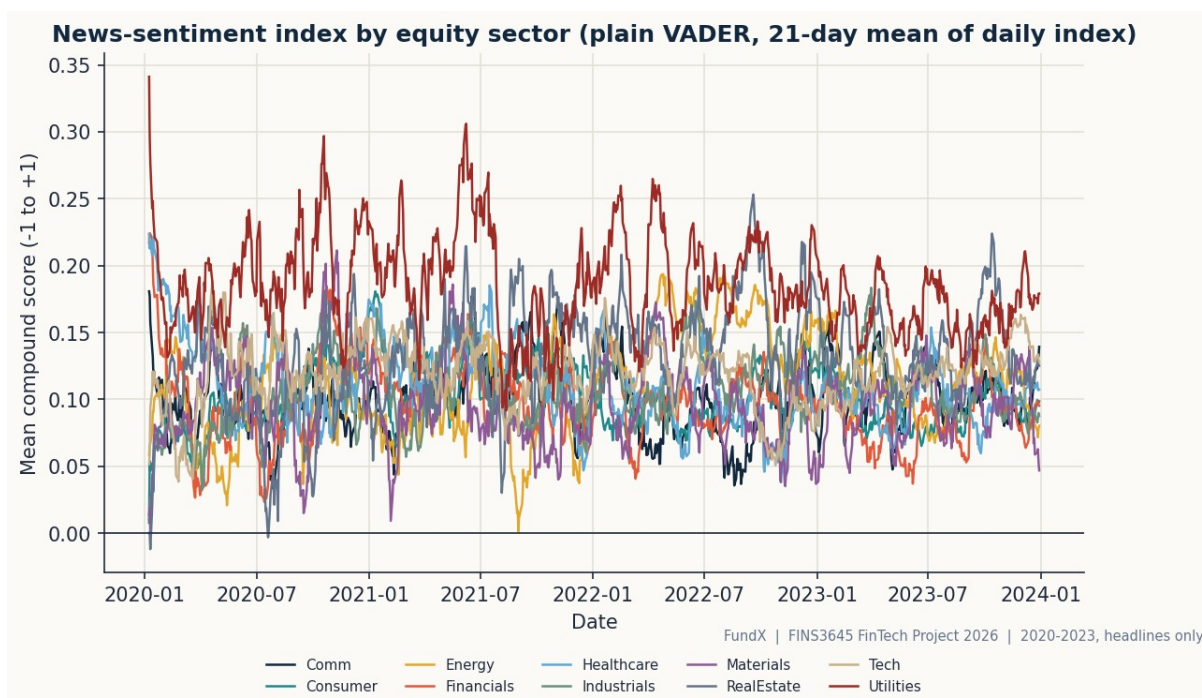


Figure 6. Equity-sector news-sentiment index: 21-day mean of the daily plain-VADER sector index. Units: mean compound score (-1 to +1), 2020-2023 headlines only.

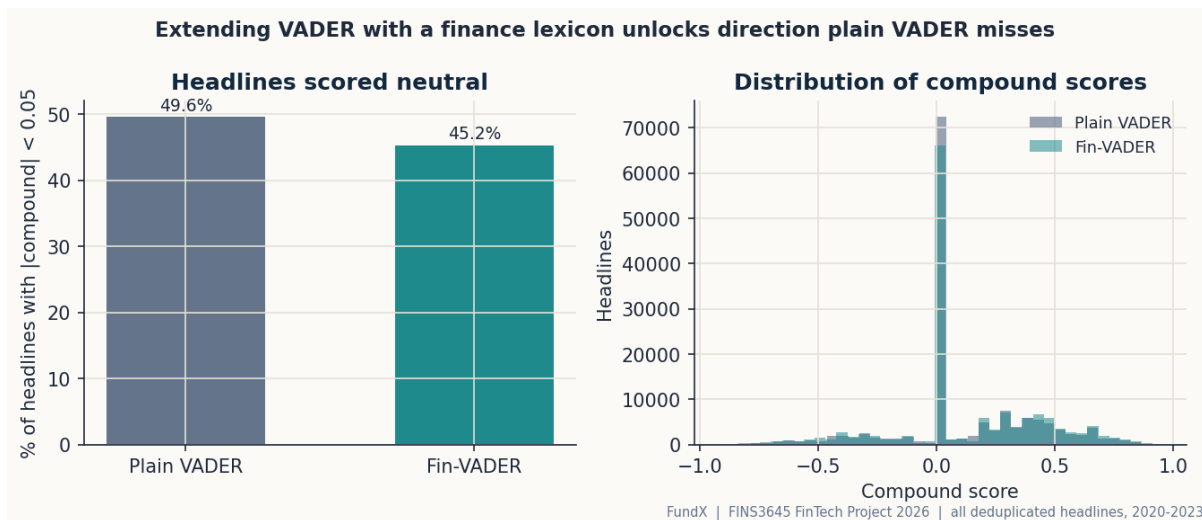


Figure 7. Fin-VADER measurement effect for all 146,836 deduplicated 2020-2023 headlines. Left: neutral share (%). Right: compound-score distribution (-1 to +1).

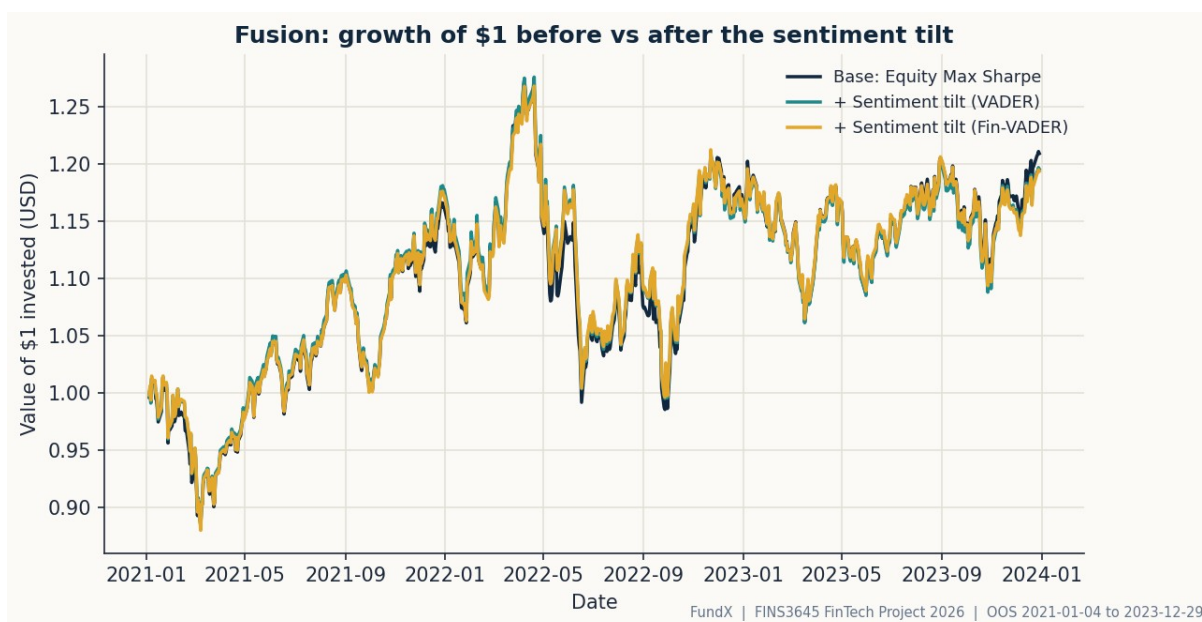


Figure 8. Growth of \$1 before and after the sentiment tilt: Equity Maximum Sharpe base versus VADER and Fin-VADER variants. Units: USD value of \$1 invested; OOS 2021-01-04 to 2023-12-29.





Figure 9. Transaction-cost model for Combined Maximum Sharpe: gross versus net-of-cost growth of \$1 using 10 bps x turnover. Units: USD value of \$1 invested; OOS 2021-01-04 to 2023-12-29.

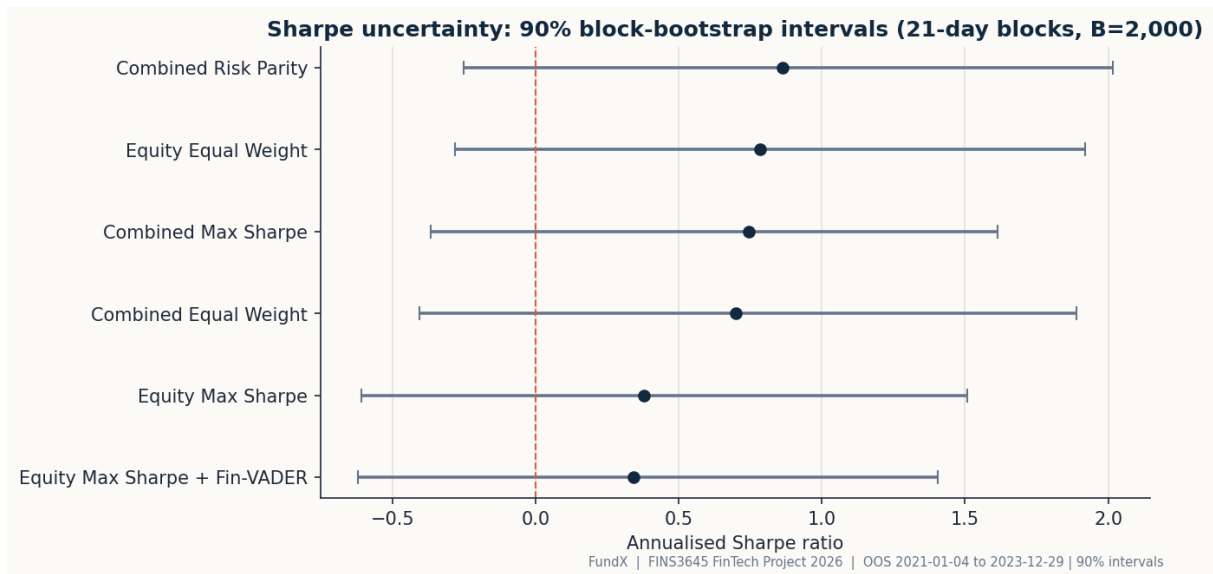


Figure 10. Annualised Sharpe ratios ( $r_f = 0$ ) with 90% moving-block-bootstrap intervals for selected funds; 21-day blocks, 2,000 draws, seed 42. OOS 2021-01-04 to 2023-12-29. Every displayed interval spans zero.